

Distributional Crowding Out Effects of Public Debt on Private Investment in Developing Economies

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Abstract

The Covid-19 pandemic, followed by financial tightening due to inflationary pressure, has raised public debt in developing economies as governments grapple with public health investments to curb the pandemic and collapse in revenues due to slower economic activity. The rise in debt may further disrupt the formal private sector in developing economies. Using two to three waves of panel firm-level data across developing economies, this study finds that higher public debt is correlated with low investment by formal private sector firms. The finding is largely driven

by small and medium-size enterprises, domestic firms, and non-exporters—raising concerns about the distributional impacts. Potential channels are uncovered. High levels of debt reduced the accessibility of finance for private sector firms, limiting investment. Furthermore, a regulatory channel is observed. As public debt rises, firms spend more time with regulatory and tax officials, which is possibly indicative of higher efforts of governments to raise revenues. This channel is stronger for small and medium-size enterprises.

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1. Introduction

The Covid-19 pandemic forced a sharp rise in global public debt as government revenue collapsed and government expenditures rose. As a result, global average general government debt rose from 84 percent of GDP in 2019 to 99 percent in 2020. In emerging market economies, average general government debt rose from 52 percent of GDP in 2019 to 64 percent in 2020 (IMF, 2021). In 2021, the share of debt in global GDP fell, due to economic recovery, but remains elevated and is projected to rise again in 2023 (IMF, 2023). Further financial tightening since the onset of the Ukraine war amid inflationary pressure has hurt government budgets even further as expenditures linked to food and energy subsidies have risen while rising interest rates have increased the challenge of rolling over or restructuring debt.

The global rise in public debt resurrects the debates about the benefits and risks of public debt-financed government spending (see Gatti et al., 2021 for a discussion). On the one hand, public debt can alleviate short-run financial constraints, allowing governments to increase or maintain public consumption and investment. Governments, especially in developing countries, have many pressing needs with potentially high returns on certain types of spending but lack the resources to finance these activities. Public debt serves

as a source of these financial resources. Public investment, financed by public debt, can directly improve the economy's productive capacity by increasing the marginal product of private capital and labor. As time progresses, this could generate positive effects on private investment and private consumption (Leduc and Wilson, 2013). Izquierdo et al. (2019) find that countries with a low initial public capital stock (as a proportion of GDP) have significantly higher public investment multipliers than countries with a high initial public capital stock. Eden and Kraay (2014) report that in low-income countries, on average, an extra dollar of government investment raises private investment by roughly two dollars and output by 1.5 dollars.

On the other hand, public debt can be problematic if it is used for unproductive ends. More precisely, if the economic and social returns to fiscal spending are lower than the realized interest rates paid for debt, the debt burden relative to national income will tend to grow over time. Even when the economic and social returns to fiscal spending exceed the interest rate on the debt used to finance it, governments still need to be cautious about borrowing more when debt is already high. As debt rises, it creates fear that governments will be unable to repay, resulting in higher risk premia and higher long-term real interest rates (Reinhart et al., 2012). The tools at the disposal of governments, such as counter-cyclical fiscal policy, are also severely weakened. Elevated debt reduces creditworthiness, restricting access to further financing and curbing the ability of economies to roll over (that is, refinance maturing) debt. High debt levels may be monetized, leading to inflation, currency depreciation, capital flight, and possibly debt or financial crises (Boskin, 2020). All of these could cause private firms to reduce investment. This channel is referred to as the “crowding out” effect of public debt. The impact of high public investment on private investment therefore remains an open empirical question.

This paper examines the impact of public debt on private firms' investment using the World Bank Enterprise Surveys, which provide a large sample of firms in many developing countries. Our contributions are threefold. First, we find that higher public debt significantly and robustly reduces private investment on the extensive and intensive margins. In other words, firms are less likely to invest, and they invest less when public debt rises. Firm panel estimations reveal that a given firm is 0.1 percent less likely to buy land and

buildings and 0.2 percent less likely to buy equipment for each 1 percent increase in central government debt. On the intensive margin, investment by a given firm drops by 2.4 percent for each 1 percent increase in central government debt. The decline is larger in equipment investment (-2.3 percent), although the drop in land building investment is quite substantial (-0.9 percent). Interestingly, we do not identify the nonlinear effect of public debt investment.

Second, we examine the distributional effects of rising public debt. In other words, investment of which types of firms is more likely to be crowded out? We find that small, fully privately owned, service, and non-exporting firms reduce investment more when public debt rises. This finding suggests that higher public debt can deepen the unequal playing field among firms. This finding is the main contribution of our paper.

Third, we examine the channels of transmission from high public debt to reduced private investment. We confirm the access to finance channel: rising public debt makes access to finance more difficult. In addition, we identify for the first time a new channel that we refer to as a regulatory channel. We discover that when public debt rises, firms spend more time with regulatory and tax officials. The activities could suggest increasing efforts of the government to collect tax in more difficult times. Interestingly, the impact of this channel is much stronger for small and fully privately-owned firms.

Our paper uses firm-panel data from the World Bank's Enterprise Surveys (ES). The standard ES sample collects information on a representative sample of formal (registered) private firms operating in manufacturing or service sectors. For countries with multiple waves of data, a subset of firms is re-interviewed over time, producing a panel component. This study exploits this panel component of about 14,000 unique forms in 57 economies between 2006 and 2019. The panel data allow for the empirical estimations to include firm-level fixed effects, thereby accounting for time-invariant firm-specific omitted variables. Section 2 discusses the data in much more detail.

Our paper is related to the discussion on the short-term need versus long-term cost of public debt. In the short-run, public debt can alleviate financial constraints, allowing governments to increase or maintain public consumption and investment. Governments, especially in developing countries, have many pressing spending needs with potentially high returns, such as the spending necessitated by the pandemic, but may lack the resources to finance these activities. Public debt provides those financial resources. Nevertheless, the short-term impact of government spending is also a matter of debate, which is related to the literature on fiscal multipliers. Friedman (1978) discussed the potential crowding out effects of financing government deficits. He argued that the magnitude of the real crowding out effect depends on whether economic resources are fully employed. The literature tends to find that the fiscal multiplier is larger (i.e., a one dollar government spending has the larger effect on the economy) when the output gap is large (Corsetti et al., 2012; Riera-Crichton et al. 2015; Auerbach and Gorodnichenko, 2012), public debt is low (Ilzetzki et al., 2013; Huidrom et al., 2020), and countries have good governance (Izquierdo et al., 2019).

In addition, government debt, while bringing short-run benefits, may incur long-run costs. Our paper is related to a vast literature on the effect of government debt and growth. The findings in this literature are ambivalent. Starting from the seminal contribution of Reinhart and Rogoff (2010), many studies have investigated this relationship, attempting to identify to what extent debt accumulation has a detrimental effect on GDP growth. The literature seems to refute the idea of a common threshold across countries (Eberhardt and Presbitero, 2015; Chudik et al., 2017). But the literature does not rule out the possibility of country-specific debt thresholds—including ones in debt intolerant countries that result in “extreme duress” if crossed (Reinhart et al., 2003 and Reinhart and Rogoff, 2009).

More directly, our paper is related to the evidence of the impact of government debt on firms’ investment, which is much more scant. Huang et al. (2020) is closest to our approach. They use Chinese firm-level data to estimate the crowding-out effect of rising public debt in China. They find that local public debt in China crowded out private firms’ investment by tightening their funding constraints while leaving state-owned firms’ investment unaffected. Moreover, the impact is stronger for small firms than larger firms, a finding

we also discover in our paper for a much broader set of firms across 57 countries. One key advantage of our study compared to Huang et al. (2020) is we could use firm fixed effects to control for unobservable firm characteristics. Huang et al. (2018), using the cross-country Orbis firm database by Bureau van Dijk, also document a negative correlation between public debt and corporate investment, especially for industries dependent on external finance. However, the Orbis data is skewed towards large publicly listed firms. Similarly, Dermici et al. (2019) use cross-country Compustat data on publicly listed firms to document a negative relationship between government debt and corporate leverage, an imperfect proxy for investment. Our paper focuses on a larger set of firms with richer information that allows us to take a step further by exploring the distribution effects of high public debt and the channels of the effects (finance and regulatory channels).

2. Data and Econometric Specifications

2.1 Data

The main source of firm-level data is the World Bank's Enterprise Surveys (ES). The standard ES sample collects information on a representative sample of formal (registered) private firms operating in manufacturing or service sectors. The ES data are fully comparable across countries and are collected via face-to-face interviews with business owners or top managers using a global standardized methodology. The survey is implemented using stratified random sampling with location (within country), sector, and size as the strata. The ES universe excludes informal firms (unregistered), fully government-owned enterprises, and micro firms (firms with fewer than five full-time employees). Several quality control checks are employed throughout the survey implementation process. Supervisors and enumerators attend formal training sessions to ensure the best practices are deployed. In addition, consistency checks are employed for 10 percent and 50 percent batches of the data during the survey to allow quick callbacks to respondents to be undertaken when necessary to verify the information.

For countries with multiple data waves, a subset of firms is re-interviewed over time, producing a panel component. This study exploits this panel component for 57 economies¹ surveyed at least twice between 2006 and 2019, amounting to approximately 14,000 unique firms. The panel data allows for the empirical estimations to include firm-level fixed effects, thereby accounting for time-invariant firm-specific omitted variables. However, there are some limitations of the panel subset of firms. First, typically less than 50 percent of firms surveyed in the first round are followed between the two waves, and thus the selection of firms may not be random and consists of survivor firms. Second, there are also high rates of attrition between the two waves.

An important feature of the ES data is the breadth of coverage in terms of topics. The surveys cover a wide range of topics, including access to finance, corruption, infrastructure, crime, competition, labor, the business environment, and performance. This enables the analyses to explore various channels of the effects, allowing for the possibility to build on studies using datasets based on registration lists (Huang et al., 2018). The data have been widely used by several studies to explore the private sector in developing economies (Paunov, 2016; Besley and Mueller, 2018; Chauvet and Ehrhar, 2018; Hjort and Poulsen, 2019).

Firm-level investment is the main outcome, captured by two types of variables. The first is a binary variable that takes a value of 1 if the firm purchased any fixed assets such as machinery, vehicles, equipment, land, or buildings were purchased (new or used) in the previous fiscal year. The second variable is the amount of expenditure on the fixed assets. These variables can be further broken down into type of investment – (a) land and buildings and (b) equipment and machinery. Summary statistics can be found in Table 1.

The main variable of interest is central government debt as a percentage of GDP at the country level. This data is obtained from the IMF Global Debt Database (Mbaye et al., 2018). The data is available for virtually the entire world. The integrity of the data has been checked through consultations with IMF country desks.

¹ See Appendix A5 for the list of countries and years.

Summary statistics can be found in Table 1. Other control variables, namely, population and real GDP per capita (constant 2010 USD), are from the World Bank's World Development Indicators.

Table 1: Summary Statistics - Panel

	Obs	Mean	Std. Dev.	Min	Max
Firm Purchased Fixed Assets Y/N	26,172	0.42	0.49	0.00	1.00
Log of Amount Investment (+1)	26,187	4.24	5.38	0.00	26.33
Log of Amount Investment in Land and Buildings (+1)	25,337	1.06	3.23	0.00	26.19
Log of Amount Investment in Equipment (+1)	25,712	4.06	5.25	0.00	26.33
Central government debt, % of GDP	26,187	39.33	28.14	0.55	336.22
Log of real GDP per capita	26,187	7.83	0.92	5.85	9.62
Log of age of firm	26,187	2.64	0.75	0.00	5.23
Log of Total Population	26,187	16.04	1.25	13.17	19.37
Log of size	26,187	2.74	1.14	-1.10	9.82
Direct exports 10% or more of sales Y/N	26,187	0.11	0.31	0.00	1.00
Foreign ownership Y/N	26,187	0.09	0.28	0.00	1.00
Establishment has checking or savings account Y/N	26,187	0.87	0.34	0.00	1.00

Source: Authors' calculation from the Enterprise Surveys.

2.2 Econometric specification

The following equation is estimated using the panel data set:

$$Invest_{it} = \beta_0 + \beta_1 GovDebt_{c,t-1} + \beta_x Z_{ct} + \beta_y Z_{it} + \gamma_1 D_i + \gamma_2 D_t + \epsilon_{it} \quad (1)$$

where *Invest* is a measure of investment by firm *i* at time *t*; *GovDebt* is the central government debt as a percentage of GDP for country *c*, lagged by 1 year²; Other control variables at the country level (*Z_{ct}*) include log of real GDP per capita and log of total population. Other control variables at the firm-level include log of age of firm, log of firm size, whether the firm exports 10 percent or more of sales (direct

² This is to capture pre-existing government debt. Using further lags of GovDebt does not change the findings.

exports), whether the firm has foreign ownership, if the firm has checking or savings account, and whether the firm is in the manufacturing sector.

The investment variable includes both a binary variable of whether the firm purchased fixed assets over the last fiscal year, and the log of investment over the last fiscal year. These variables can be further broken down into investment in land and buildings as well as machinery and equipment. Equation (1) is estimated using OLS with firm (D_i) and year (D_t) fixed effects. The estimation strategy exploits within firm variation over time. We attenuate the possibility of reverse causality by lagging the Debt variable by 1 year. It is also unlikely that investment by a single firm is likely to affect the aggregate debt held by the central government in an economy. Omitted variable bias is an important concern. The use of firm fixed effects accounts for time-invariant firm-level unobservable variables. Thus, if an unobserved characteristic of a firm, such as its political connectivity, does not change over time, it is absorbed into the firm-level fixed effects. The year fixed effects account for global shocks such as a financial crisis. In addition, we account for a host of variables that are based on the conceptual underpinnings of the relationships and the literature.

The level of development of the economy and the size of the economy are captured by the log of real GDP per capita and the log of total population. The size and development of the economy affects the size of the markets and the ability of the private sector to invest. It may also capture the level of debt the government can take on.

To guard against further omitted variable bias, we account for a number of control variables at the firm level. Firm size can influence the level of investment as larger firms may have different investment needs and may also have more access to resources and networks to carry out investments than smaller firms (Cull and Xu, 2005). Thus, we account for firm size. Several studies have uncovered relationships between a firm's performance and its age and size (Biesebroek 2005; Bigsten and Gebreyesus 2007; Haltiwanger et al. 2013). Technology upgrades are likely to be influenced by the age of the firm (Cull and Xu, 2005). Thus, we account for the age of the firm in our estimations. Outward orientation, such as exporter status and foreign ownership, of the firm is correlated with innovation and productivity, and therefore, with

potentially higher level of investment (Seker, 2012; Lopez 2005; Bernard et al. 2007; Dimelis and Louri, 2002; Guadalupe et al., 2012). Therefore we control for variables such as exporter status and foreign ownership. Access to finance influences investment decisions and has also improved firm productivity (Gatti and Love, 2008; Rajan and Zingales, 1998). We account for this by whether or not a firm has a checking of savings account. Summary statistics for all these variables can be found in Table 1.

3. Empirical Findings

3.1 Does public debt crowd out private investment?

We start by examining the extensive margin of investment, namely, firms deciding to invest or not. Table 2 shows that when public debt rises, a firm is much less likely to invest. A firm is 0.1 percent less likely to buy land and buildings and 0.2 percent less likely to buy equipment for each 1 percent increase in central government debt (see columns [1] and [2] of Table 2). Note that firm fixed effects are controlled for. Hence, we only examine the effect within a firm and control for unobservable firm characteristics.

Table 2: Public Debt and The Extensive Margin of Investment

Panel	Firm Purchased Land and Buildings Y/N	Firm Purchased Equipment Y/N
	(1)	(2)
Central government debt, % of GDP, lagged (t-1)	-0.001** (0.000)	-0.002*** (0.001)
Log of real GDP per capita	-0.339* (0.203)	-0.438 (0.344)
Log of age of firm	-0.008 (0.013)	-0.012 (0.029)
Log of Total Population	0.050 (0.249)	-0.157 (0.398)
Log of firm size	0.023 (0.015)	0.077*** (0.026)

Direct exports 10% or more of sales Y/N	0.048 (0.031)	0.083 (0.052)
Foreign ownership Y/N	0.005 (0.030)	-0.006 (0.056)
Establishment has checking or savings account Y/N	-0.018 (0.022)	0.038 (0.038)
Manufacturing Sector Y/N	0.004 (0.035)	0.018 (0.063)
Constant	1.901 (4.019)	6.321 (7.138)
Year Fixed Effects	YES	YES
Firm Fixed Effects	YES	YES
Number of observations	25,337	25,712
Adjusted R2	0.364	0.401

Note: *** p<0.01, ** p<0.05, * p<0.1. Regressions use survey weights. Standard errors are clustered at the country level

Next, we examine the intensive margin of investment, namely, the amount of a firm's investment when public debt rises. The dependent variable is log of investment in current USD.³ To avoid taking log of zero investment, we add one unit (USD 1) to the investment variable. Table 3 reveals that when public debt rises, firms invest significantly less. Real investment by firms drops by 2.4 percent for each 1 percentage increase in central government debt. The decline is larger in equipment investment (-2.3%), although the drop in land building investment is quite substantial (-0.9%). Note that firm fixed effects are included in the estimations.

Table 3: Public Debt and The Intensive Margin of Investment

Panel	Log of Amount Investment (+1)		
	Total Investment	Land and Building	Equipment
Central Gov debt, % of GDP (t-1)	-0.024*** (0.006)	-0.009*** (0.003)	-0.023*** (0.005)
Log of real GDP per capita	-2.851 (3.791)	-2.896* (1.714)	-2.371 (3.759)
Log of age of firm	-0.098 (0.313)	-0.040 (0.130)	-0.098 (0.295)

³ To the extent that the year fixed effects capture the USD inflation, the use of current USD is equivalent to real USD.

Log of Total Population	1.264 (4.084)	0.158 (2.052)	0.892 (4.274)
Log of size	1.071*** (0.247)	0.289** (0.147)	1.017*** (0.252)
Direct exports 10% or more of sales Y/N	0.914 (0.567)	0.504 (0.327)	0.852 (0.560)
Foreign ownership Y/N	0.059 (0.558)	-0.006 (0.335)	0.180 (0.563)
Establishment has checking or savings account Y/N	0.368 (0.387)	-0.145 (0.196)	0.383 (0.373)
Manufacturing Sector Y/N	0.114 (0.623)	0.048 (0.353)	0.207 (0.602)
Constant	5.039 (75.836)	20.348 (34.677)	7.338 (81.230)
Firm Fixed effects	YES	YES	YES
Year Fixed effects	YES	YES	YES
Number of observations	26,187	25,337	25,712
Number of unique firms	14,160	14,041	14,080
Adjusted R2	0.429	0.352	0.437

Note: *** p<0.01, ** p<0.05, * p<0.1. Regressions use survey weights. Standard errors are clustered at the country level

The debt literature has considered the possibility of threshold of effects of public debt (see a seminal paper by Reinhart and Rogoff (2010)). We therefore consider a non-linear effect of public debt on private investment by using a quadratic term of public debt. We fail to see a significant non-linear effect of public debt. The results are shown in Appendix Table A1.

3.2 Distributional effects of rising public debt

An important contribution of our paper is to examine the impact on different kinds of firms. We find that investments of small, service, non-exporting, and domestic firms are affected more when public debt rises. For the first time, the distributional effects of rising debts are systematically reported. Small and medium-

size firms are defined as those having 5 to 99 employees.⁴ Non-exporters are defined as those whose export revenue is less than 10 percent of sales. Table 4 reports that investment of a small firm drops by 2.3 percent for each 1 percent increase in central government debt. The declines are 3 percent for service firms, 2.4 percent for non-exporter firms, and 2.5 for domestic firms. The estimated coefficients for larger firms, manufacturing firms, exporting firms, and foreign-owned firms are either smaller or comparable to those by other firms. Still, in all cases, the coefficients are less precisely estimated, indicating a less statistically significant impact of rising debt on these firms. In addition, the distributional effects are stronger for investment in equipment than for land and buildings (see Appendix Tables A2 and A3).

It is important to note that the effects nested the potentially positive effects of government spending associated with the rising debt. In other words, the effects captured here are the net effects. Appendix Table A4 presents the findings for the universe of firms and without firm fixed effects. Again, the findings remain very robust.

3.3 On the channels of impacts

Understanding the channel by which higher public debt reduces a firm's investment is important for policy makers. The evidence can guide governments' support for firms when they have to increase public debt. The literature has well established the "access to finance" channel. Higher debt makes access to finance for the private sector more difficult. Hence firms are forced to reduce investment (see Huang et al., 2018 and 2020). We also find evidence for the access to finance channel in the following way. When government debt is higher, firms are more likely to rely on internal funds for working capital and investment (see columns [1] and [2] of Table 5). Working capital from other sources (such as microfinance institutions, credit cooperatives, credit unions, finance companies, moneylenders, friends, relatives) is significantly

⁴ Recall that firms with fewer than 5 workers are not included in the Enterprise Surveys.

reduced. Again, it is interesting that the shift from other sources to internal funds is more significant for small and medium-size firms.

Table 4: The Distributional Effects of Rising Public Debt

Log of Amount Investment (+1)	SME	Large	Manufacturing	Services	Exporters	Non- exporters	Foreign- owned (10 pct or more)	Domestic Owned
	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Central government debt, % of GDP, lagged (t-1)	-0.023*** (0.006)	0.023 (0.036)	-0.009 (0.007)	-0.030*** (0.006)	-0.028 (0.050)	-0.024*** (0.006)	-0.012 (0.017)	-0.025*** (0.007)
Constant	YES	YES	YES	YES	YES	YES	YES	YES
Year Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES
Firm Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES
Other controls	YES	YES	YES	YES	YES	YES	YES	YES
Number of observations	21,437	4,750	14,298	11,889	4,002	22,185	2,477	23,710
Adjusted R2	0.383	0.569	0.482	0.410	0.524	0.418	0.567	0.425

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at the country level. Small and medium firms (SME) are defined as those having 5 to 99 employees. Non-exporters are defined as those whose export revenue is less than 10 percent of sales. Log of real GDP per capita, Log of age of firm, Log of Total Population, Log of size, Direct exports, Foreign ownership, checking or savings account, sector are controlled for. Survey weights are used.

Table 5: The Access to Finance Channel

Panel A: Working Capital

	Working Capital - Internal Funds (%)		Working Capital - Banks (%)		Working Capital - Supplier Credit (%)		Working Capital - Others (%)	
Sample	SME	Large	SME	Large	SME	Large	SME	Large
Central government debt, % of GDP, lagged (t-1)	0.046*	-0.006	-0.010	0.078	0.016	0.011	-0.052***	-0.083*
	(0.024)	(0.132)	(0.017)	(0.127)	(0.018)	(0.069)	(0.009)	(0.047)
Constant	YES	YES	YES	YES	YES	YES	YES	YES
Firm Fixed effects	YES	YES	YES	YES	YES	YES	YES	YES
Year Fixed effects	YES	YES	YES	YES	YES	YES	YES	YES
Other controls	YES	YES	YES	YES	YES	YES	YES	YES
Number of observations	20,225	4,559	20,225	4,559	20,225	4,559	20,225	4,559
Adjusted R2	0.410	0.639	0.458	0.623	0.375	0.546	0.327	0.664

Panel B: Investment

	Investment - Internal Funds (%)		Investment - Banks (%)		Investment - Supplier Credit (%)		Investment - Equity or Stock Sales (%)		Investment - Other (%)	
Sample	SME	Large	SME	Large	SME	Large	SME	Large	SME	Large
Central government debt, % of GDP, lagged (t-1)	0.041	-0.065	0.046	0.047	-0.002	0.031	-0.026	-0.017	-0.060**	0.005
	(0.080)	(0.229)	(0.051)	(0.159)	(0.022)	(0.047)	(0.031)	(0.047)	(0.028)	(0.181)
Constant	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Firm Fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year Fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Other controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Number of observations	8,811	3,332	8,811	3,332	8,811	3,332	8,811	3,332	8,811	3,332
Adjusted R2	0.399	0.558	0.417	0.487	0.186	0.475	0.226	0.449	0.493	0.607

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at the country level. Small and medium firms (SME) are defined as those having 5 to 99 employees Log of real GDP per capita, Log of age of firm, Log of Total Population, Log of size, Direct exports, Foreign ownership, checking or savings account, sector are controlled. Other sources include non-bank financial institutions such as microfinance institutions, credit cooperatives, credit unions, finance companies, moneylenders, friends, relatives, etc. (%). Survey weights are used.

We also document a novel channel not previously examined in the literature. The channel is referred to as “the regulatory channel”. In particular, we find that when government debt is higher, firms are more likely to receive visits from tax officials and spend more time dealing with government regulations requirements (see Table 6). This is a channel that can potentially hurt firm operations. Again, the effects are much stronger and more statistically significant for small and medium-size firms. This disproportionate effect can exacerbate the unequal playing field against small and medium-size firms. We do not have further evidence regarding the motivation of regulatory enforcement. However, we speculate that government officials face more pressure to raise taxes and other revenue as government debt rises.

Table 6: The Regulatory Channel

Model	OLS with Firm and Year FE					
Outcome Variable	Senior management time spent in dealing with requirements of government regulations (%)			Average number of visits or required meetings with tax officials		
Sample	All	SME	Large	All	SME	Large
Central government debt, % of GDP, lagged (t-1)	0.019 (0.017)	0.025* (0.014)	0.014 (0.055)	0.008*** (0.003)	0.007** (0.003)	0.004 (0.008)
Constant	YES	YES	YES	YES	YES	YES
Firm Fixed effects	YES	YES	YES	YES	YES	YES
Year Fixed effects	YES	YES	YES	YES	YES	YES
Other controls	YES	YES	YES	YES	YES	YES
Number of observations	24,789	20,156	4,633	26,268	21,463	4,805
Adjusted R2	0.360	0.338	0.498	0.444	0.449	0.488

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at the country level. Log of real GDP per capita, Log of age of firm, Log of Total Population, Log of size, Direct exports, Foreign ownership, checking or savings account, sector are controlled for. Survey weights are used.

4. Summary and Conclusions

The Covid-19 pandemic and the ensuing financial tightening due to the inflationary environment has raised public debt as governments grapple with public health investments to curb the pandemic and collapse in revenues due to slower economic activity. The rise in debt may further disrupt the formal private sector in developing economies. Using two to three waves of panel firm-level data across developing economies, this study finds that higher public debt is correlated with low investment by formal private sector firms. The finding is largely driven by small and medium-size enterprises, domestic firms, and non-exporters –

raising concerns about the distributional impacts. But why do those firms suffer more? Those firms may have weaker access to finance and hence are more likely to have their investment crowded out, as shown in Section 3. It is well established that small and domestic firms have more difficult access to finance – so the finding is consistent with our findings. In addition, we are the first to identify that these more vulnerable firms can face stricter enforcement as public debt rises.

What implications do these findings have? First, we document that public debt crowds out private investment, the level effects. Second, we also document that rising public debt deepens firms' inequality, the distributional effects. The findings hence carry important policy implications: on average, in developing countries, rising debt hurts investment; and more importantly, it hurts small, service, domestic firms more, deepening the already unequal playing field. Therefore, measures should be strongly considered to support vulnerable firms during periods of rising debt, such as during and after the pandemic. This may include improving the business environment for firms to find opportunities to grow. And possibly provide access to finance during difficult times as witnessed during the pandemic.

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Appendix Table A1: Non-linear effect of public debt

Outcome Variable	Log of Amount Investment (+1)		
Sample	All	SME	Large
	(1)	(2)	(3)
Central government debt, % of GDP, lagged (t-1)	-0.032 (0.026)	-0.035 (0.028)	0.054 (0.086)
Central government debt Square, lagged (t-1)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.001)
Log of real GDP per capita	-3.257 (3.494)	-3.630 (3.791)	-0.229 (4.394)
Log of age of firm	-0.092 (0.313)	-0.129 (0.339)	0.119 (1.057)
Log of Total Population	1.193 (3.745)	1.339 (4.001)	9.368 (11.520)
Log of size	1.068*** (0.245)	0.975*** (0.280)	1.505 (1.165)
Direct exports 10% or more of sales Y/N	0.911 (0.565)	0.941 (0.663)	0.853 (1.502)
Foreign ownership Y/N	0.061 (0.558)	-0.136 (0.637)	1.360 (1.102)
Establishment has checking or savings account Y/N	0.364 (0.391)	0.377 (0.382)	1.242 (1.980)
Manufacturing Sector Y/N	0.116 (0.623)	-0.098 (0.709)	2.689 (2.056)
Constant	9.717 (65.160)	10.590 (69.135)	-152.645 (191.574)
Firm Fixed effects	YES	YES	YES
Year Fixed effects	YES	YES	YES
Number of observations	26,187	21,437	4,750
Adjusted R2	0.429	0.383	0.569

note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at the country level. Survey weights are used.

Appendix Table A2: Investment in Equipment

Log of Amount Investment in Equipment (+1)	SME	Large	Manufacturing	Services	Exporters	Non- exporters	Foreign- owned (10 pct or more)	Domestic Owned
	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
Central government debt, % of GDP, lagged (t-1)	-0.022*** (0.006)	0.027 (0.034)	-0.007 (0.007)	-0.028*** (0.006)	-0.035 (0.049)	-0.023*** (0.006)	-0.023 (0.017)	-0.023*** (0.006)
Constant	YES	YES	YES	YES	YES	YES	YES	YES
Year Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES
Firm Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES
Other controls	YES	YES	YES	YES	YES	YES	YES	YES
Number of observations	21,097	4,615	14,048	11,664	3,924	21,788	2,404	23,308
Adjusted R2	0.391	0.580	0.491	0.418	0.540	0.423	0.538	0.434

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at the country level. Log of real GDP per capita, Log of age of firm, Log of Total Population, Log of size, Direct exports, Foreign ownership, checking or savings account, sector are controlled for. Survey weights are used.

Appendix Table A3: Investment in Land and Building

Log of Amount Investment in Land and Building (+1)	SME	Large	Manufacturing	Services	Exporters	Non- exporters	Foreign- owned (10 pct or more)	Domestic Owned
Central government debt, % of GDP, lagged (t-1)	-0.009*** (0.003)	0.005 (0.031)	-0.002 (0.007)	-0.010*** (0.003)	-0.035 (0.049)	-0.023*** (0.006)	-0.002 (0.009)	-0.009*** (0.003)
Constant	YES	YES	YES	YES	YES	YES	YES	YES
Year Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES
Firm Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES
Other controls	YES	YES	YES	YES	YES	YES	YES	YES
Number of observations	20,813	4,524	13,799	11,538	3,924	21,788	2,387	22,950
Adjusted R2	0.359	0.370	0.402	0.354	0.540	0.423	0.428	0.356

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at the country level. Log of real GDP per capita, Log of age of firm, Log of Total Population, Log of size, Direct exports, Foreign ownership, checking or savings account, sector are controlled for. Survey weights are used.

Appendix Table A4: Public Debt and Investment – Cross-Sectional Global Sample

Sample	ALL	SME	Large	Manufacturing	Services	Exporters	Non-exporters	Foreign-owned	Domestic Owned
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Central government debt, % of GDP, lagged (t-1)	-0.010** (0.004)	-0.010** (0.004)	-0.011 (0.010)	-0.012*** (0.003)	-0.010** (0.005)	-0.011 (0.013)	-0.009** (0.004)	-0.009 (0.009)	-0.010*** (0.004)
Log of real GDP per capita	-2.080 (1.270)	-2.191* (1.238)	-1.156 (2.435)	-2.809* (1.472)	-2.107 (1.320)	0.143 (2.186)	-2.161* (1.295)	-0.918 (2.249)	-2.199* (1.318)
Log of age of firm	-0.292*** (0.046)	-0.326*** (0.048)	-0.040 (0.167)	-0.422*** (0.064)	-0.243*** (0.056)	-0.111 (0.147)	-0.318*** (0.048)	-0.077 (0.146)	-0.314*** (0.052)
Log of Total Population	1.760 (2.293)	1.467 (2.126)	2.706 (4.870)	2.825 (2.849)	0.842 (2.150)	-1.840 (3.056)	2.290 (2.281)	-1.160 (3.513)	1.978 (2.389)
Log of size	1.395*** (0.056)	1.474*** (0.063)	1.326*** (0.127)	1.299*** (0.069)	1.463*** (0.063)	1.300*** (0.092)	1.425*** (0.057)	1.451*** (0.115)	1.379*** (0.056)
Direct exports 10% or more of sales Y/N	0.453*** (0.159)	0.555*** (0.188)	0.440* (0.259)	0.239 (0.168)	0.763*** (0.238)			0.126 (0.257)	0.531*** (0.173)
Foreign ownership Y/N	0.032 (0.112)	0.002 (0.128)	0.324 (0.288)	0.130 (0.224)	-0.042 (0.158)	-0.170 (0.207)	0.080 (0.127)		
Establishment has checking or savings account Y/N	0.800*** (0.120)	0.818*** (0.120)	0.830** (0.421)	0.853*** (0.140)	0.804*** (0.136)	0.368 (0.425)	0.839*** (0.123)	1.274*** (0.308)	0.759*** (0.125)
Manufacturing Sector Y/N	0.068 (0.080)	0.107 (0.085)	-0.335 (0.252)			-0.293 (0.260)	0.131* (0.079)	-0.053 (0.267)	0.088 (0.087)
Constant	-13.126 (43.749)	-7.446 (40.856)	-37.105 (89.473)	-24.320 (54.853)	2.022 (40.384)	30.893 (64.896)	-21.531 (42.999)	24.874 (65.691)	-15.567 (45.953)
Year Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
Country Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
Number of observations	89,672	73,068	16,604	49,025	40,647	12,938	76,734	9,350	80,322
Adjusted R2	0.167	0.134	0.181	0.193	0.156	0.185	0.159	0.190	0.162

Note: This Table shows the results when all Enterprise Survey firms are used (as opposed to the baseline regression where the sample is restricted to panel data where firms appear at least twice). Hence in this Table, country fixed effects are included in place of firm fixed effects. *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at the country level. Survey weights are used.

Appendix Table A5: Country and Year Composition of Sample

Country	Survey Year	Country	Survey Year
Albania	2013	Indonesia	2009
Albania	2019	Indonesia	2015
Argentina	2006	Jordan	2013
Argentina	2010	Jordan	2019
Argentina	2017	Kazakhstan	2009
Armenia	2009	Kazakhstan	2013
Armenia	2013	Kazakhstan	2019
Azerbaijan	2009	Kenya	2007
Azerbaijan	2013	Kenya	2013
Belarus	2008	Kenya	2018
Belarus	2013	Kosovo	2013
Belarus	2018	Kosovo	2019
Benin	2009	Kyrgyz Republic	2009
Benin	2016	Kyrgyz Republic	2013
Bhutan	2009	Kyrgyz Republic	2019
Bhutan	2015	Lao PDR	2009
Bolivia	2006	Lao PDR	2012
Bolivia	2010	Lao PDR	2016
Bolivia	2017	Lao PDR	2018
Bosnia and Herzegovina	2009	Lesotho	2009
Bosnia and Herzegovina	2013	Lesotho	2016
Bosnia and Herzegovina	2019	Liberia	2009
Bulgaria	2009	Liberia	2017
Bulgaria	2013	Mali	2007
Bulgaria	2019	Mali	2010
Cameroon	2009	Mali	2016
Cameroon	2016	Mexico	2006
Chad	2009	Mexico	2010
Chad	2018	Moldova	2009
Colombia	2006	Moldova	2013
Colombia	2010	Moldova	2019
Colombia	2017	Mongolia	2009
Côte d'Ivoire	2009	Mongolia	2013
Côte d'Ivoire	2016	Mongolia	2019
Ecuador	2006	Montenegro	2009
Ecuador	2010	Montenegro	2013
Ecuador	2017	Montenegro	2019
Guatemala	2006	Morocco	2013
Guatemala	2010	Morocco	2019
Guatemala	2017		
Honduras	2006		
Honduras	2010		
Honduras	2016		